If a car accelerating from zero takes just 8 seconds to travel 200 m, its average velocity for the 8 second interval is $\frac{200}{8} = 25$ m/s. The Mean Value Theorem says that at some point during the travel the speedometer must read exactly 90 km/h which is equal to 25 m/s.

Theorem 7.4

If $f(x)$ is continuous in closed interval $[a,b]$ and differentiable in open interval (a,b) and if $f'(x) > 0,\ \forall x \in (a,b)$, then for, $x_1, x_2 \in [a,b]$, such that $x_1 < x_2$ we have, $f(x_1) < f(x_2)$.

Proof

By the mean value theorem, there exists a $c \in (x_1, x_2) \subset (a,b)$ such that,

$$\frac{f(x_2) - f(x_1)}{x_2 - x_1} = f'(c)$$

Since $f'(c) > 0$, and $x_2 - x_1 > 0$ we have $f(x_2) - f(x_1) > 0$.

We conclude that, whenever $x_1 < x_2$, we have $f(x_1) < f(x_2)$.

Remark

If $f'(x) < 0,\ \forall x \in (a,b)$, then for, $x_1, x_2 \in [a,b]$, such that $x_1 < x_2$ we have, $f(x_1) > f(x_2)$.

The proof is similar.

Example 7.25

Find the values in the interval $(1,2)$ of the mean value theorem satisfied by the function $f(x) = x - x^2$ for $1 \le x \le 2$.

Solution

$f(1) = 0$ and $f(2) = -2$. Clearly $f(x)$ is defined and differentiable in $1 < x < 2$. Therefore, by the Mean Value Theorem, there exists a $c \in (1,2)$ such that

$$f'(c) = \frac{f(2) - f(1)}{2 - 1} = 1 - 2c$$

That is, $1 - 2c = -2 \Rightarrow c = \frac{3}{2}$.

Geometrical meaning

Geometrically, the mean value theorem says the secant to the curve $y = f(x)$ between $x = a$ and $x = b$ is parallel to a tangent line of the curve, at some point $c \in (a,b)$.

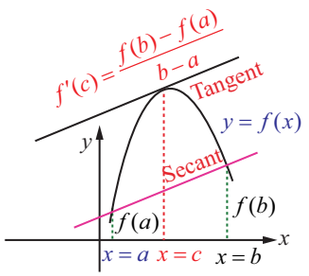


Fig.7.14

Consequences of Lagrange's Mean Value Theorem

There are three important consequences of MVT for derivatives.

(1) To determine the monotonicity of the given function (Theorem 7.4)

(2) If $f'(x) = 0$ for all x in (a,b), then f is constant on (a,b).

(3) If $f'(x) = g'(x)$ for all x, then $f(x) = g(x) + C$ for some constant C.

Chapter 7 Differential Calculus Original-new.indd 19 26-02-2020 17:23:01